

SOLVING DYNAMIC MODELS WITH AGGREGATE SHOCKS AND HETEROGENEOUS AGENTS

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I develop an algorithm for solving dynamic models in which individual decision rules and the cross-sectional distribution of agents' characteristics influence each other. To illustrate the algorithm, I solve an endowment economy with incomplete markets, a continuum of heterogeneous agents, and aggregate shocks. The key innovation of the algorithm is to parameterize the (cross-sectional) density with a flexible functional form, which makes it possible to avoid simulation techniques. The paper shows how to check for accuracy and establishes links between the properties of the incomplete-markets economy and the aspects involved in obtaining a numerical solution.

Keywords: Dynamic Models, Aggregate Shocks, Heterogeneous Agents

1. INTRODUCTION

There are many examples in macroeconomics in which an individual's decision depends not only on individual specific variables but also on the characteristics of the other agents in the economy.¹ If some of these characteristics are endogenous and the number of agents is large, then it is hard to solve the model because the solution is a function of a high-dimensional set of time-varying state variables. I develop an algorithm to solve these types of models. To illustrate the algorithm, I solve an endowment economy with incomplete markets, a continuum of heterogeneous agents, and aggregate shocks. In this economy, the cross-sectional distribution of asset holdings and income levels has predictive power for future interest rates. Therefore, individual decisions depend on this cross-sectional distribution, which is itself an endogenous variable that varies over time.

Other papers in the literature deal with the problem of a high-dimensional set of endogenous state variables in one of three ways. The first way is to assume an uncountable number of agents and abstract from aggregate uncertainty.² If agents are, *ex ante*, the same but differ in the realizations of the shock they receive, a law of

The author would like to thank James Bullard, Marco Cagetti, Lars Hansen, Ken Judd, Tony Smith, Hal White, participants of the Stanford Institute for Theoretical Economics workshop, and an anonymous referee for useful comments. This paper was prepared for the conference "Computation and Estimation in Finance and Economics" at Washington University in St. Louis. Address correspondence to: Wouter J. den Haan, Department of Economics, University of California, San Diego, La Jolla, CA 92093-0508, USA; e-mail: wdenhaan@ucsd.edu.

large numbers can be invoked to show that the cross-sectional distribution is time invariant. The second approach considered in the literature is to limit the number of agents to two.³ As a result, the cross-sectional distribution of asset holdings and income levels can be characterized by the aggregate levels, and the individual levels of one agent. The third method is to use a linear approximation.⁴ All three approaches have their limitations. The assumptions of the first approach imply that asset prices and other aggregate variables are constant, an undesirable feature for many purposes. With the second approach, individual specific shocks have aggregate effects because there are only two agents.⁵ Finally, the third approach can be used only when nonlinearities are not important, which usually is not the case in incomplete-market economies. In these types of models, nonlinearities are consequential because of borrowing constraints and because of the large amount of idiosyncratic risk that typically is assumed.

Den Haan (1996a) and Krusell and Smith (1994) develop algorithms that avoid making these assumptions. Both papers start in on the problem of a high-dimensional set of endogenous state variables by approximating the cross-sectional distribution using moments. Both papers obtain information about the functional form of the cross-sectional distribution using simulation techniques. The key innovation of the algorithm proposed in this paper is to parameterize the (cross-sectional) density with a flexible functional form, which makes it possible to avoid simulation techniques. This reduces the computational burden of obtaining accurate solutions substantially, because simulation techniques are subject to sampling errors that disappear at a slow rate.⁶ To improve the efficiency and the convergence properties, the algorithm incorporates several of the techniques proposed by Christiano and Fisher (1994) and Judd (1991, 1992).

The paper also establishes a link between the properties of incomplete-market models and the aspects involved in numerically solving these models. In particular, I discuss nonlinearities in the demand for bonds as a function of the wealth level as well as nonlinearities in the relationship between the bond price and the amount of cross-sectional wealth dispersion. Finally, I document nonlinearities in the response of the bond price to an aggregate shock. This type of discussion should be helpful for researchers interested in a quantitative analysis of models with incomplete markets and heterogeneous agents.

The outline of the paper is as follows: In Section 2, a simple model with incomplete markets is discussed. In Section 3, I discuss the solution algorithm. In Section 4, I show how to check the accuracy of a solution and that it is feasible to obtain accurate solutions. Section 4 also includes a discussion of the properties of the incomplete-markets economy and the challenges they create for the algorithm. The last section concludes.

2. AN INCOMPLETE-MARKETS ECONOMY

To illustrate the algorithm, I solve an exchange economy with incomplete markets and a continuum of heterogeneous agents. Recently, models with incomplete

markets have been used successfully to study asset prices and consumption behavior [cf. Aiyagari and Gertler (1991), Deaton (1991), Telmer (1993), Lucas (1994), and Heaton and Lucas (1996)]. Except for the presence of both aggregate uncertainty and a large number of agents, the model described in this section is very similar to the ones used in the literature. Den Haan (1996b) documents the differences and similarities between this model and the corresponding model with only two agents. In Section 2.1, I describe the baseline version of the model and in Section 2.2, I discuss the parameter values for which I solve the model.

2.1. Model

The model is a standard incomplete-markets economy. Each agent lives forever and maximizes his expected utility. Agents are ex ante the same but differ ex post, because they receive different realizations from the stochastic endowment process. In particular, agent j 's endowment relative to the per-capita endowment, y_t^j , is a first-order Markov process. The per-capita endowment is equal to A_t and agent j 's income is equal to $y_t^j A_t$. I use lowercase to denote variables that are growth rates or variables that are relative to the per-capita endowment and uppercase to denote variables that are in levels. Agent j receives either the high-income endowment, in which case y_t^j is equal to θ_h^y , or the low-income endowment, in which case y_t^j is equal to θ_l^y . The probability of moving from state e to e' is equal to $\pi_{e',e}^y$ for $e \in \{l, h\}$ and $e' \in \{l, h\}$. I assume that $\pi_{h,h}^y = \pi_{l,l}^y$. Because there are a continuum number of agents, this implies an ergodic distribution in which the fraction of high-income agents is equal to the fraction of low-income agents. The (gross) growth rate of the per-capita endowment a_t also is modeled as a stationary first-order Markov process. In particular, I assume that there are two states: the boom, in which the growth rate of the per-capita endowment is equal to θ_b^a ; and the recession, in which the growth rate equals θ_r^a . The probability of moving from state n to n' is equal to $\pi_{n',n}^a$ for $n \in \{r, b\}$ and $n' \in \{r, b\}$. I assume that $\theta_b^a - \theta_r^a > 0$, so that the per-capita endowment level is a random-walk process with positive drift.

Markets are incomplete, and agent j can smooth consumption C_t^j only by trading in a one-period bond with a risk-free payoff. Let q_t be the price of a bond at period t that delivers one unit of consumption in the next period, and let B_t^j be the amount of bonds that is purchased in period t by agent j . I assume that there is a limit on the amount that agents can go short in bonds. In particular, $B_t^j > -\bar{B}_t = -\bar{b}A_t$, where $\bar{b}$ is a positive number.

The current-period utility function is the standard constant-relative-risk-aversion utility function with a coefficient of relative risk aversion equal to γ . That is,

$$U(C_t^j) = \frac{(C_t^j)^{1-\gamma} - 1}{1-\gamma}, \quad \text{when } \gamma \neq 1,$$

and

$$U(C_t^j) = \ln C_t^j, \quad \text{when } \gamma = 1.$$

Agent j 's maximization problem thus can be written as follows:

$$\max_{\{C_t^j, B_t^j\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(C_t^j),$$

subject to

$$C_t^j + q_t B_t^j = y_t^j A_t + B_{t-1}^j, \quad (1)$$

$$B_t^j \geq -\bar{b} A_t, \quad \text{and} \quad B_{-1}^j \text{ given.} \quad (2)$$

The first-order conditions for the agent's problem are given by equations (1), (2), and the following two-part Kuhn–Tucker condition:

$$q_t [C_t^j]^{-\gamma} \geq \beta E_t [C_{t+1}^j]^{-\gamma}, \quad (3)$$

$$(B_t^j + \bar{b}) [q_t (C_t^j)^{-\gamma} - \beta E_t (C_{t+1}^j)^{-\gamma}] = 0.$$

The last three equations easily can be transformed to a system of equations that contains only stationary variables. To see this, define $c_t^j = C_t^j / A_t$ and $b_t^j = B_t^j / A_t$. Equations (1), (2), and (3) then can be written as

$$c_t^j + q_t b_t^j = y_t^j + b_{t-1}^j / a_t, \quad (4)$$

$$b_t^j \geq -\bar{b}, \quad (5)$$

$$q_t [c_t^j]^{-\gamma} \geq \beta E_t [c_{t+1}^j a_{t+1}]^{-\gamma}, \quad \text{and} \quad (6)$$

$$(b_t^j + \bar{b}) [q_t (c_t^j)^{-\gamma} - \beta E_t (c_{t+1}^j a_{t+1})^{-\gamma}] = 0.$$

Let w_t^j stand for the beginning-of-period wealth of agent j , i.e., $w_t^j = y_t^j + (b_{t-1}^j) / a_t$. In the economy with a continuum of agents, the state variables of agent j are y_t^j , w_t^j , and the aggregate state variables. The aggregate state variables are a_t and the joint distribution of endowment levels and bondholdings. Because there are only two possible realizations for the endowment shock, the joint distribution is completely characterized by the cumulative (cross-sectional) distribution function of beginning-of-period bondholdings of the agents who receive the low-income shock, F_t^l , and the cumulative (cross-sectional) distribution function of beginning-of-period bondholdings of the agents who receive the high-income shock, F_t^h .

A solution to the model consists of a consumption function $c(w_t^j, y_t^j, a_t, F_t^l, F_t^h)$, an investment function $b(w_t^j, y_t^j, a_t, F_t^l, F_t^h)$, a bond price function $q(a_t, F_t^l, F_t^h)$, and the functionals $F^L(a_t, a_{t-1}, F_{t-1}^l, F_{t-1}^h)$ and $F^H(a_t, a_{t-1}, F_{t-1}^l, F_{t-1}^h)$ that describe the law of motion of F_t^l and F_t^h , respectively. The only reason why the current value of a_t is an argument of the transition functionals is that the beginning-of-period bondholdings are scaled relative to the per-capita endowment. For example, suppose that in period $t - 1$ an agent borrows the maximum amount $\bar{b} A_{t-1}$. Thus,

$b_{t-1}^l = -\bar{b}$. This means that in period t , beginning-of-period bondholdings, relative to the aggregate endowment, are equal to $-b/a_t$. This example also makes clear that when the borrowing constraint is binding, F_t^l and F_t^h have point mass at $-b/a_t$. The functions $c(\cdot)$, $b(\cdot)$, $q(\cdot)$, and the functionals $F^l(\cdot)$ and $F^h(\cdot)$ must satisfy equations (4) through (6) and the equilibrium condition

$$\int_{-\bar{b}/a_t}^{\infty} b(\theta_l^y + \zeta, \theta_l^y, a_t, F_t^l, F_t^h) dF_t^l(\zeta) + \int_{-\bar{b}/a_t}^{\infty} b(\theta_h^y + \zeta, \theta_h^y, a_t, F_t^l, F_t^h) dF_t^h(\zeta) = 0. \quad (7)$$

In addition, the functionals $F^l(\cdot)$ and $F^h(\cdot)$ describe the law of motion of the cross-sectional dispersion of bondholdings. In particular, if the investment decision is given by $b(\cdot)$ and agents' bondholdings are distributed according to F_t^l and F_t^h , then next period's cross-sectional distribution of bondholdings is given by $F^l(a_{t+1}, a_t, F_t^l, F_t^h)$ and $F^h(a_{t+1}, a_t, F_t^l, F_t^h)$ for the agents who receive the low-income draw and the high-income draw, respectively.

Several papers in the literature discuss the existence of equilibrium. Existence of an equilibrium in an economy like ours but with a finite number of agents is discussed by Magill and Quinzii (1994). In Section 4, I compare the numerical solution to the model having a continuum of agents with the simulated results of an economy having 100,000 agents and conclude that the results are very similar. This suggests that there is no fundamental difference between the economy with a continuum of agents and the economy with a large but finite number of agents. Related papers are Levine (1989), Levine and Zame (1993), and Duffie et al. (1994).

2.2. Parameter Values

Heaton and Lucas (1996) obtained parameter estimates for the stochastic driving processes using the income series from the Panel Studies of Income Dynamics (PSID) under the assumption that both processes are two-state first-order Markov processes. The two values for the idiosyncratic shock, y_t , are 0.7544 and 1.2456, and the two values for the aggregate (gross) growth rate, a_t , are 0.991 and 1.0470. The probability of receiving in this period the same idiosyncratic shock as last period is equal to 0.7412 and the probability of receiving the same aggregate shock is equal to 0.5473. In the baseline version of the model, however, I set the transition probability of the idiosyncratic shock equal to 0.5, which implies that idiosyncratic shocks are serially uncorrelated. The advantage of doing this is that the cross-sectional distribution of the low-income agents is the same as the cross-sectional distribution of the high-income agents. This choice, thus, reduces the dimension of the set of state variables and the computational costs of the algorithm. Although most of the analysis uses the baseline version of the model,

TABLE 1. Parameter values

Parameter	Baseline model	Alternative parameters considered
θ_b^a	1.047	—
θ_r^a	0.991	—
$\pi_{b,b}^a = \pi_{r,r}^a$	0.5473	0.7
$\theta_h^y (= 2 - \theta_l^y)$	1.2456	—
$\pi_{h,h}^y = \pi_{l,l}^y$	0.5	0.7412
β	0.965	—
$\underline{\gamma}$	3.0	—
$\underline{b}$	1.0	0.2

I also consider the case in which the idiosyncratic shocks are serially correlated and the set of state variables includes the distribution of bondholdings for the low-income types as well as the distribution for the high-income types. The time period in the model corresponds to a year and the discount rate is set equal to 0.965. The only remaining parameters are the degree of relative risk aversion, γ , and the borrowing-constraint parameter, $\underline{b}$. I have solved the model using values for γ equal to -1 , -3 , and -5 and using values for $\underline{b}$ ranging from 0.2 to 2.0. Without losing any additional insights, I only report the results for the case in which γ is equal to 3 and when $\underline{b}$ is equal to 1 and 0.2. The parameter values used in the baseline model, as well as the alternative parameters considered, are presented in Table 1.

3. SOLUTION ALGORITHM

In this section, I discuss the solution algorithm. In Section 3.1, I give an overview of the algorithm. In Section 3.2, I discuss the approach used to approximate the cross-sectional distribution. In Section 3.3, I discuss the general method of function approximation. In Section 3.4, I give the specific details on the approximating functions. In Section 3.5, I discuss the disadvantage of simulation techniques for this problem and discuss a modification of the algorithm.

3.1. Overview of Algorithm

In the model developed in Section 2, the bond price is a function of the growth rate of the aggregate endowment a and the cross-sectional distribution of bondholdings. A key feature of the proposed algorithm is to approximate the cross-sectional distribution of income and bondholdings using a finite $M \times 1$ vector ϕ , containing moments of the cross-sectional distribution.⁷ To approximate the bond price function, I therefore use the function $\Theta(a, \phi; \delta^\Theta)$, where δ^Θ is a vector of parameters and $\Theta(\cdot)$ is chosen from a class of functions that can approximate any function arbitrarily well. Similarly, I use the function $\Phi(a', a, \phi; \delta^\Phi)$ to approximate the

transition law of ϕ . Because ϕ is a vector, $\Phi(\cdot)$ is a vector-valued function. Solving the individual problem requires approximating one more function. Standard choices are the value function, the policy rule for consumption, or the conditional expectation in the Euler equation. I approximate the conditional expectation and denote the approximating function by $\Psi(w, y, a, \phi; \delta^\Psi)$.⁸ Also, let s be the vector $[w, y, a, \phi]$, $\bar{s}$ be the vector $[a, \phi]$ and $\bar{s}'$ be the vector $[a', \bar{s}]$.

For a particular functional form for $\Theta(\cdot)$, $\Phi(\cdot)$, and $\Psi(\cdot)$, the algorithm solves for the parameter values δ^Φ , δ^Θ , and δ^Ψ with the following iteration scheme:

Step 1. Given parameter values for δ^Φ and δ^Θ , solve the individual's problem. That is, obtain parameter values for δ^Ψ .

Step 2. Given the decision rules for the individuals, solve the aggregate problem. That is, obtain parameter values for δ^Φ and δ^Θ . If these parameter values are close to those selected to solve the individual's problem in step 1, the algorithm has converged. If not, then repeat steps 1 and 2.

These steps are discussed in Section 3.3. Obtaining solutions to this problem is feasible in a reasonable amount of computing time. For example, when the solution for the case with $b = 0.4$ is used as the initial guess for the case where $b = 0.2$, it takes the algorithm, written in FORTRAN approximately 1 h on a 486DX-66MHz personal computer to find a solution.

3.2. Approximating the Cross-Sectional Distribution

The difficulty in solving models like the one discussed in Section 2 is that the solution depends on an infinite dimensional object, the cross-sectional distribution of income levels and bondholdings. The key feature of the proposed algorithm is that the cross-sectional distribution of income and bondholdings is approximated with a flexible functional form with a finite number of coefficients. In particular, I use an element from the class of exponentials. Because of the presence of borrowing constraints, I adjust the class of functions to allow for the possibility of having positive probability mass at the borrowing constraints. In particular, for $e \in \{l, h\}$, I use the following cumulative distribution function as the n th-order approximation for the distribution of the bondholdings of the agents who receive the income shock e ,

$$\begin{aligned} P^e(b) &= 0 & b < \bar{b}/a, \\ P^e(b) &= \rho_0^e \int_{-\infty}^b \exp(\rho_1^e \zeta + \rho_2^e \zeta^2 + \cdots + \rho_n^e \zeta^n) d\zeta & b \geq \bar{b}/a. \end{aligned} \quad (8)$$

For this approximation, there are $2n + 1$ aggregate state variables, namely a , the n coefficients of the distribution of the bondholdings of the low-income agents, and the n coefficients of the distribution of the high-income agents.⁹ In this paper, I follow Den Haan (1996a) and use the moments instead of the coefficients of the distribution as the state variables. Given the chosen functional form for the density, the first n moments capture the same information as the $n + 1$ coefficients of the density.

To find the coefficients of the distribution that correspond to a set of moments, one must solve a nonlinear problem. Suppose, for example, that two moments are used to approximate the cross-sectional distribution. To find the values of ρ_0^l , ρ_1^l , and ρ_2^l that correspond to particular values of the mean ϕ_1 , and to the variance of the distribution of bondholdings of the low-income agents ϕ_2 , a nonlinear equation solver is used to solve the following set of equations.¹⁰

$$\begin{aligned} 1 &= \rho_0^l \int_{-\infty}^{\infty} \exp(\rho_1^l b + \rho_2^l b^2) db, \\ \phi_1 &= \rho_0^l \int_{-\infty}^{\infty} \max(b, \bar{b}/a) \exp(\rho_1^l b + \rho_2^l b^2) db, \quad \text{and} \\ \phi_2 &= \rho_0^l \int_{-\infty}^{\infty} [\max(b, \bar{b}/a) - \phi_1]^2 \exp(\rho_1^l b + \rho_2^l b^2) db. \end{aligned} \quad (9)$$

As part of the solution procedure, the researcher must specify a grid for the moments of the distribution. At each grid point, one has to solve for the corresponding parameters of the distribution. Especially for higher-order approximations, this is not a trivial exercise for the following two reasons: First, the problem is highly nonlinear and the equation solver will only find a solution for good initial conditions. Second and more important, there is no guarantee that all combinations of moments considered are consistent with a finite-order exponential distribution.¹¹ This illustrates a disadvantage of using rectangular grids. Rectangular grids include points that may not make sense at all, such as a combination of a high value for the second moment and a low value for the fourth moment. In some cases, I have, therefore, adjusted the grid. The details are discussed in Section 3.4.

3.3. General Method of Function Approximation

In this section, I discuss the class of functions used to approximate the unknown functions and the method used to obtain the coefficients of the approximating functions $\Psi(\mathbf{s}, \delta^\Psi)$, $\Theta(\bar{\mathbf{s}}, \delta^\Theta)$, and $\Phi(\bar{\mathbf{s}}, \delta^\Phi)$.¹² These approximating functions are chosen from the class of Chebyshev polynomials. To solve for the coefficients of the approximating functions, I minimize the sum of squared approximation errors across the elements of a grid. Chebyshev roots are used to construct the grid. Because Chebyshev roots take on values between -1 and 1 , I use a linear operator to transform the state variables so that the lower bound is equal to -1 and the upper bound is equal to 1 . For simplicity, I do not introduce new notation for the transformed variables. Let δ_i^j denote the value of δ^j at the i th iteration. The following three-step iteration scheme can be used to find the coefficients:

Step 1. At each grid point, calculate the object of interest. For example, when solving for the coefficients of $\Psi(\cdot)$, calculate $E\{[c(s')a']^\gamma \mid \mathbf{s}\}$; when calculating the coefficients

of $\Theta(\cdot)$, calculate the bond price. Let K^j be the total number of grid points and let $\mathbf{Y}^j$ denote the $K^j \times 1$ vector of the calculated values for $j \in \{\Psi, \Theta, \Phi\}$.

Step 2. Obtain updated values for the coefficients δ^j , δ_{new}^j by regressing the values of $\mathbf{Y}^j$ on the Chebyshev polynomial terms. Let X^j be the $K^j \times N^j$ matrix where the k th row is equal to the N^j Chebyshev terms at the k th grid point. The formula for δ_{new}^j is given by

$$\delta_{\text{new}}^j = (X^{j'} X^j)^{-1} X^{j'} \mathbf{Y}^j, \quad j \in \{\Psi, \Theta, \Phi\}. \quad (10)$$

Note that because of the discrete orthogonality property of Chebyshev polynomials, the matrix $(X^{j'} X^j)^{-1}$ is diagonal, which means that calculating the inverse is trivial.

Step 3. Use the method of successive approximation to update δ , that is,

$$\delta_{i+1}^j = \nu \delta_{\text{new}}^j + (1 - \nu) \delta_i^j, \quad (11)$$

where the adjustment parameter ν is between zero and one. Choosing the value for ν usually requires some experimentation. A higher value of ν has the potential benefit of faster convergence but also carries the risk of global instability.

In the remainder of this section, I discuss the details of the first step for the three different functions. I start with the individual's problem. Given a bond price function $\Theta(\cdot)$ and a transition function $\Phi(\cdot)$, the individual's problem consists of finding a consumption function $c(s)$ and an investment function $b(s)$ that satisfy, for all possible values of s , the following three equations:

$$c(s) + \Theta(\bar{s})b(s) = y + b_{-1}/a, \quad (12)$$

$$b(s) \geq -\bar{b}, \quad (13)$$

and

$$\Theta(\bar{s})[c(s)]^{-\gamma} \geq \beta E(\{c[y' + b(s)/a', y', a', \Phi(\bar{s})]a'\}^{-\gamma} | s), \quad (14)$$

$$[b(s) + \bar{b}][\Theta(\bar{s})[c(s)]^{-\gamma} - E(\{c[y' + b(s)/a', y', a', \Phi(\bar{s})]a'\}^{-\gamma} | s)] = 0.$$

Because ϕ is an $M \times 1$ vector, the number of state variables is equal to $M + 3$. The key step in the algorithm used here is to replace the conditional expectation in equation (14) by the polynomial $\Psi(w, y, a, \phi; \delta^\Psi)$. At each grid point, I calculate the conditional expectation of next period's marginal utility $[c(s')a']^{-\gamma}$, where $s' = [y' + b(s)/a', y', a', \phi]$. I use $\Phi(\cdot; \delta^\Phi)$ to calculate ϕ' and I use equations (12) through (14), $\Psi(s; \delta_i^\Psi)$, and $\Theta(\bar{s}; \delta^\Theta)$ to calculate $b(s)$. Next period's marginal utility depends on two random variables that can take on two values. Calculating the conditional expectation requires, therefore, calculating the weighted sum of only four values.¹³ I use equations (12) through (14), $\Psi(\cdot; \delta_i^\Psi)$, and $\Theta(\cdot; \delta^\Theta)$ to calculate $c(s')$ for the four possible realizations of s' . Denote the calculated value of $E[c(s')a']^{-\gamma}$ at the i th grid point by Y_i^Ψ and let Y_i^Ψ comprise the i th element of the $K \times 1$ vector $\mathbf{Y}^\Psi$.

At each grid point, the bond price in the first step of the aggregate problem to find the coefficients δ^Θ is calculated as follows: First, use equations 9 to find the

parameters of the distribution of the bondholdings that correspond with the value of ϕ . Next, find the bond price as the value for which the aggregate demand for bonds is equal to zero. That is, find the bond price q that solves the following equation:

$$\int_{-\infty}^{\infty} b(\theta_l^y + \zeta, \theta_l^y, \bar{s}; q, \delta^\Psi) dP^l(\zeta) + \int_{-\infty}^{\infty} b(\theta_h^y + \zeta, \theta_h^y, \bar{s}; q, \delta^\Psi) dP^h(\zeta) = 0. \quad (15)$$

The investment function $b(s; q, \delta^\Psi)$ is implicitly defined in equations (12) through (14) with the conditional expectation replaced by $\Psi(s; \delta^\Psi)$. To evaluate the integrals in equation (15), Simpson or Hermite–Gaussian quadrature is used. Let K^Θ be the number of grid points and let the calculated bond price at the i th grid point comprise the i th element of the $K^\Theta \times 1$ vector Y^Θ .

Finally, I describe the first step of the problem to solve for the coefficients of the transition function $\Phi(a', a, \phi_1, \phi_2, \phi_3; \delta^\Phi)$. Recall that the only reason why a' is an argument of $\Phi(\cdot)$ is that beginning-of-period bondholdings are defined relative to the per-capita endowment. The algorithm first calculates the moments relative to last period's per-capita endowment, $\bar{\Phi}(a, \phi_1, \phi_2, \phi_3; \delta^\Phi)$. The elements of $\Phi(\cdot)$ then can be found by dividing the elements of $\bar{\Phi}(\cdot)$ by a^k , where k is the order of the particular moment. At each grid point, the moments of next period's cross-sectional distribution of bondholdings relative to this period's per-capita endowment are calculated. Suppose that three moments are used to characterize the cross-sectional distribution and that ϕ_1 is the average amount of bondholdings of the agents who receive the low-income shock, ϕ_2 is the variance of the bondholdings of the agents who receive the low-income shock, and ϕ_3 is the variance of the bondholdings of the agents who receive the high-income shock. Let $\bar{\phi}_1'$, $\bar{\phi}_2'$, and $\bar{\phi}_3'$ be the corresponding values in the next period relative to the current per-capita endowment. These are calculated using the following equations:

$$\begin{aligned} \bar{\phi}_1' &= \pi_{l,l} \int_{-\infty}^{\infty} b(\theta_l^y + \zeta, \theta_l^y, a, \phi_1, \phi_2, \phi_3; q, \delta^\Psi) dP^l(\zeta) \\ &\quad + \pi_{l,h} \int_{-\infty}^{\infty} b(\theta_h^y + \zeta, \theta_h^y, a, \phi_1, \phi_2, \phi_3; q, \delta^\Psi) dP^h(\zeta), \\ \bar{\phi}_2' &= \pi_{l,l} \int_{-\infty}^{\infty} [b(\theta_l^y + \zeta, \theta_l^y, a, \phi_1, \phi_2, \phi_3; q, \delta^\Psi) - \bar{\phi}_1']^2 dP^l(\zeta) \\ &\quad + \pi_{l,h} \int_{-\infty}^{\infty} [b(\theta_h^y + \zeta, \theta_h^y, a, \phi_1, \phi_2, \phi_3; q, \delta^\Psi) - \bar{\phi}_1']^2 dP^h(\zeta), \quad \text{and} \\ \bar{\phi}_3' &= \pi_{l,l} \int_{-\infty}^{\infty} [b(\theta_l^y + \zeta, \theta_l^y, a, \phi_1, \phi_2, \phi_3; q, \delta^\Psi) + \bar{\phi}_1']^2 dP^l(\zeta) \\ &\quad + \pi_{l,h} \int_{-\infty}^{\infty} [b(\theta_h^y + \zeta, \theta_h^y, a, \phi_1, \phi_2, \phi_3; q, \delta^\Psi) + \bar{\phi}_1']^2 dP^h(\zeta), \end{aligned} \quad (16)$$

where $q = \Theta(a, \phi_1, \phi_2, \phi_3; \delta^\Theta)$. Again, equations (9) are used to solve for the parameters of the exponential density that correspond with the moments ϕ_1 , ϕ_2 , and ϕ_3 , and a quadrature procedure is used to evaluate the integrals.

Recall from the discussion in Section 3.1 that the three problems are not solved simultaneously, but that the algorithm iterates between the individual problem and the aggregate problem. That is, when the algorithm solves for the coefficients δ^Ψ , it takes the bond price and the transition function as given. Because the grid for the aggregate problems is a subset of the grid for the individual problem, one could just use the values calculated at the grid points instead of using the approximating functions $\Theta(\cdot)$ and $\Phi(\cdot)$. In fact, the algorithm does not require an approximating functional form for the transition function. In contrast, one does need a functional form for the bond price, because calculating the conditional expectation also requires knowledge of next period's bond price¹⁴ and next period's state is not necessarily a grid point.

3.4. Specific Details

In this section, I provide some details about the numerical solutions. The settings most commonly used in this paper are labeled EXP2Q500 and EXP4Q101. Details are presented in Table 2. To construct the grid space, one has to choose upper and lower bounds for the values of b_{-1}/a as well as for the elements of ϕ . The borrowing-constraint parameter gives a lower bound for the level of bondholdings. The upper bound is set equal to four times the value of the lower bound.¹⁵ Initial guesses for the bounds of the elements of ϕ can be found by solving the model without aggregate uncertainty. Unfortunately, some additional trial and error is usually required before sensible bounds are found.

Note that EXP4Q101 does not use the average amount of bondholdings of the low-income types and therefore, can be used only to solve models in which the idiosyncratic shock is an i.i.d. r.v. EXP2Q500 can be used to solve models in which the idiosyncratic shock is serially correlated. For models in which there is no serial correlation, EXP2Q500 also is used but is more expensive than necessary because in this case only the variance of the bondholdings of the low-income types is needed for a second-order approximation and the realization of the endowment shock has no additional information when the wealth level is included. As mentioned in Section 3.2, I sometimes had problems finding the coefficients of the exponential used to approximate the cross-sectional distribution for a given set of moments. This problem could be eliminated by not using a rectangular grid. Because these problems never occurred when I used a second-order approximation, I only adjust the grid when I use a fourth-order approximation. In particular, I delete those elements from the grid where the highest (lowest) value of one moment is combined with the lowest (highest) or second-to-lowest (second-to-highest) value of another moment. In the polynomial $\Psi(\cdot)$, the coefficients relating to the cross products of the cross-sectional distribution and the individual state variables are set equal to zero. Other numerical solutions that are discussed in this paper are EXP2Q30,

TABLE 2. Details of two numerical solutions^a

Property	EXP2Q500	EXP4Q101
Order of P	Second	Fourth
Moments used	Average for low-income types Variance for low-income types Variance for high-income types	Variance Skewness Kurtosis
Method of integration	Hermite–Gaussian quadrature	Simpson quadrature
No. of quadrature nodes	500	101
Dimensions grid for individual problem	$K_w = 7, K_y = 2, K_a = 2,$ $K_{\phi 1} = 5, K_{\phi 2} = 5, K_{\phi 3} = 5$	$K_w = 7, K_y = 2, K_a = 2,$ $K_{\phi 1} = 5, K_{\phi 2} = 5, K_{\phi 3} = 5$
Dimensions grid for aggregate problem	$K_a = 2, K_{\phi 1} = 5, K_{\phi 2} = 5,$ $K_{\phi 3} = 5$	$K_a = 2, K_{\phi 1} = 5,$ $K_{\phi 2} = 5, K_{\phi 3} = 5$ plus adjustment
Order of Ψ	$N_w = 7, N_y = 2, N_a = 2,$ $N_{\phi 1} = 2, N_{\phi 2} = 2,$ $N_{\phi 3} = 2$, and some zero restrictions	$N_w = 7, N_y = 2, N_a = 2,$ $N_{\phi 1} = 2, N_{\phi 2} = 2,$ $N_{\phi 3} = 2$, and some zero restrictions
Order of Θ	$N_a = 2, N_{\phi 1} = 2,$ $N_{\phi 2} = 2, N_{\phi 3} = 2$	$N_a = 2, N_{\phi 1} = 2,$ $N_{\phi 2} = 2, N_{\phi 3} = 2$
Order of Φ	$N_a = 2, N_{\phi 1} = 2,$ $N_{\phi 2} = 2, N_{\phi 3} = 2$	$N_a = 2, N_{\phi 1} = 2,$ $N_{\phi 2} = 2, N_{\phi 3} = 2$
Number of coefficients	130	130

^aThis table reports details of the approximating functions and the grid space. N_x is the highest order of variable x that is included in the approximating function and K_x is the number of Chebyshev roots used for variable x . P is the approximation of the cross-sectional distribution, Ψ is the approximation of the conditional expectation, Θ is the approximation of the bond price, and Φ is the approximation of the transition function.

which is identical to EXP2Q500 but uses only 30 Hermite quadrature nodes; EXP4Q101*, which is identical to EXP4Q101 but uses up to 15th-order polynomial terms for wealth in the parameterization of the conditional expectations.

3.5. Alternatives

In this section, I discuss alternatives to the choices made in this paper. The main difference between the algorithm developed in this paper and the algorithm developed in Den Haan (1996a) is that a flexible functional form is used to parameterize the cross-sectional distribution, which makes it possible to avoid simulation methods. A disadvantage of Monte Carlo integration techniques is that, because of the presence of sampling variation, the rate of convergence is slow.

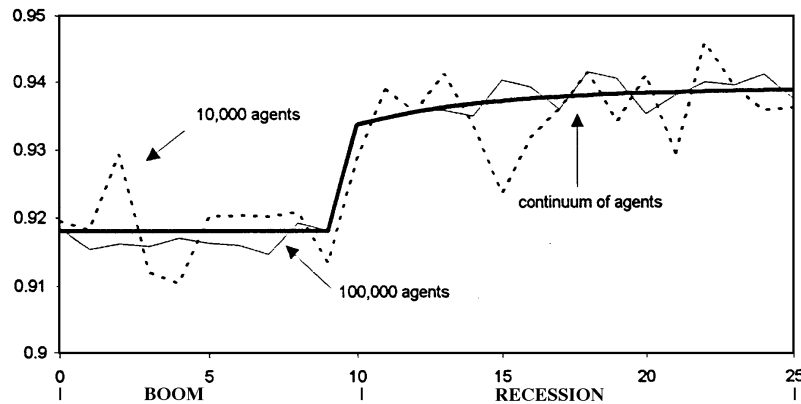


FIGURE 1. Equilibrium bond price using a finite set of agents. [This figure plots the bond price calculated using a fixed group of agents. The bond price is solved from equations (4) through (6) with the conditional expectation replaced by its numerical solution and the equilibrium condition that the demand for bonds equals zero. The parameters of the baseline version of the model are used. The numerical solution used to generate the individual policy functions is solution EXP4Q101. Details are provided in Section 3.4.]

To document the importance of sampling variation in this particular example, I perform the following experiment: Because the economy consists of a continuum of agents, the cross-sectional distribution should not display any time variation when the economy has been in a recession for a long time period. The importance of sampling variation is illustrated by calculating the equilibrium bond price using a simulation with 10,000, 100,000, and a continuum of agents. In particular, the bond price is solved using equations (4) through (6) with the conditional expectation replaced by a numerical solution¹⁶ and the equilibrium condition that the demand for bonds equals zero. The results are reported in Figure 1. First consider the time path of the bond price when a continuum of agents is used. When the economy goes into a recession, there is a sudden increase in the bond price followed by gradual further increases. The gradual adjustment that takes place after the first period is 27% of the total price change. As demonstrated in Figure 1, there is a substantial amount of sampling variation even when the number of agents is as high as 10,000. With this much noise, it would take many time-series observations to accurately describe the response of the interest rate to a persistent aggregate shock. The graph also documents that second-order statistics, such as the variance of the change in the interest rate, depend crucially on the number of agents selected to numerically find the solution. This suggests that obtaining accurate solutions of this model using the Monte Carlo techniques of Den Haan (1996a) or Krusell and Smith (1994) requires a large number of agents, which would make the algorithm computationally very expensive.

An interesting modification of the algorithm proposed here would be to use the coefficients of the cross-sectional density as the state variables instead of the

moments. The main reason for using the moments is that, given the work done in Den Haan (1996a), it was not hard to find sensible lower and upper bounds for the grid. The transition equation then can be calculated as follows: Let $p^e(\cdot)$ be the current-period density of bondholdings of agents with an endowment equal to e and let $p^{e'}(\cdot)$ be next period's density of bondholdings. The transition is given by the following equation:

$$p^{e'}(b') = \pi_{e',l} p^l [b^{-1}(b', \theta_l^y, \bar{s})] + \pi_{e',h} p^h [b^{-1}(b', \theta_h^y, \bar{s})], \quad (17)$$

where $b^{-1}(\cdot)$ is the function that indicates the current-period level of bondholdings for which an agent chooses b' . Suppose that the approximating density has two free coefficients, ρ_1^e and ρ_2^e . In this case, only two values of b' are needed, at each grid point, to determine the coefficients of $p'(\cdot)$. Let $Y^{\rho_1^l}$ be the vector containing the calculated values $Y^{\rho_1^l}$ across the grid points. Then, the transition function for $Y^{\rho_1^l}$ can be found using the procedure described in Section 3.3. I think that the approach using moments as state variables is easier to extend to cases in which the set of state variables consists of multivariate cross-sectional distributions.

4. ACCURACY

In this section, I discuss the accuracy of the numerical algorithm proposed in Section 3 for several versions of the model discussed in Section 2. The purpose of this section is threefold. First, I will discuss how to check for accuracy. Second, I will use a graphical approach to illustrate the strengths and weaknesses of the algorithm. The relative merits of an algorithm depend, of course, on the particular model being solved. The third objective of this section, therefore, is to illustrate important properties of the solution to the incomplete-markets economy discussed in Section 2 and to discuss the best way to capture these features using numerical approximations. Section 4.1 discusses the accuracy of the individual policy rules, taking the bond price and the transition functions as given. Section 4.2 discusses the accuracy of the bond price function and the transition functions, taking the functional form of the cross-sectional distribution as given. Section 4.3 discusses the accuracy of the numerical integration procedures used. Section 4.4 discusses the accuracy of the finite-order exponential used to approximate the cross-sectional distribution of bondholdings. These four sections show that several features of the model can only be calculated accurately when computer-intensive approximations are used. In Section 4.5, I show, however, that several of the first- and second-moment properties of the model that economists are usually interested in can be calculated accurately using a less computer-intensive approach.

4.1. Accuracy of Individual Policy Rules

In this section, I discuss the accuracy of the individual policy functions. Recall that an approximate solution is obtained by replacing the conditional expectation

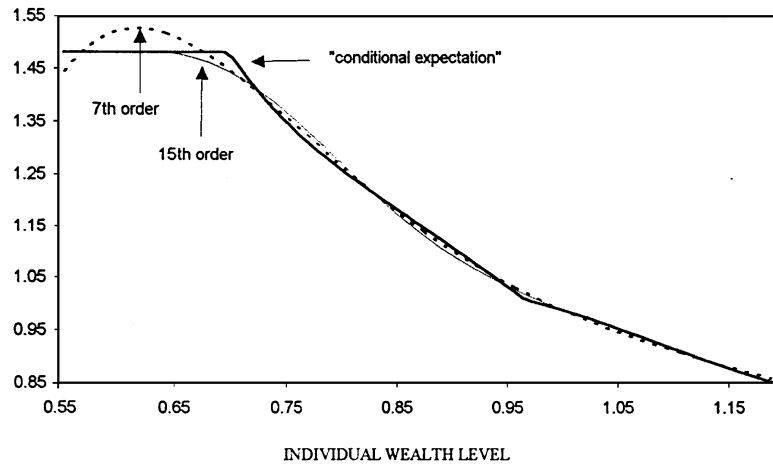


FIGURE 2. Approximations of conditional expectation. [This graph plots the 7th- and 15th-order approximation $\Psi(s, \delta^\Psi)$ and the conditional expectation of next period's marginal utility of consumption. The parameter of relative risk aversion is equal to 3 and borrowing-constraint parameter is equal to 0.2. The solution using the 7th-order polynomial corresponds with solution EXP4Q101 and the solution using the 15th-order polynomial corresponds with solution EXP4Q101*. Both are described in more detail in Section 3.4. The conditional expectation of next period's marginal utility of consumption is calculated using the policy function for consumption and bondholdings that are implied by the 15th-order approximation.]

$E\{[c(s')a']^{-\gamma} | s\}$ in equation (14) by the approximation $\Psi(s, \delta^\Psi)$. The parameters δ^Ψ are found by minimizing the distance between $\Psi(s, \delta^\Psi)$ and the conditional expectation of $E\{[c_\Psi(s')a']^{-\gamma} | s\}$ at a fixed set of grid points, where $c_\Psi(\cdot)$ is the consumption function that is implied by $\Psi(\cdot, \delta^\Psi)$. The question arises as to how good is the approximation $\Psi(s, \delta^\Psi)$ at other points in the state space. Most interesting in this respect are variations in the agent's wealth level, because in the presence of borrowing constraints the conditional expectation is not differentiable with respect to the wealth level.¹⁷ To illustrate the importance of this nondifferentiability, I focus on the case with a tight borrowing constraint.¹⁸ Note that $E\{[c_\Psi(s')a']^{-\gamma} | s\}$ is not necessarily equal to $E\{[c(s')a']^{-\gamma} | s\}$, where $c(\cdot)$ is the rational-expectations solution for the consumption function. However, when the approximating function $\Psi(s, \delta^\Psi)$ is for all choices of s equal to the function being approximated— $E\{[c_\Psi(s')a']^{-\gamma} | s\}$, then one has found the true rational-expectation solution.

I check the accuracy of numerical solutions EXP4Q101 and EXP4Q101* which use a 7th- and a 15th-order polynomial respectively, in the agent's wealth level. Figure 2 compares the approximate functions and the corresponding functions being approximated. To avoid unnecessary clutter, I only plot the conditional expectation of next period's marginal utility using the consumption function that is

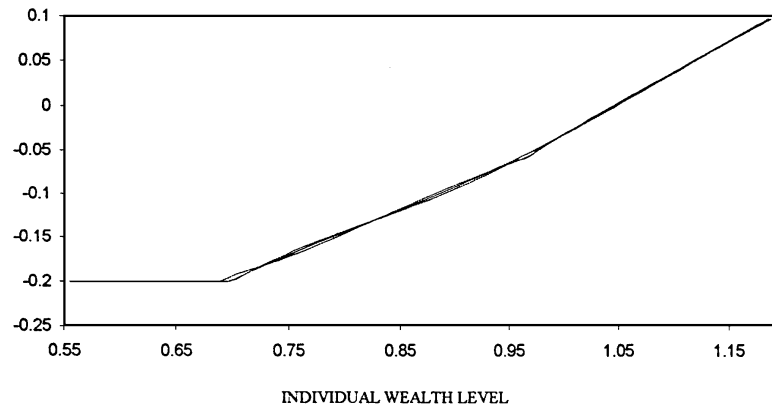


FIGURE 3. Approximations of the savings decision. [This graph plots the approximating policy functions for the investment in bonds that is implied by a 7th- and 15th-order approximation for the conditional expectation of next period's marginal utility. The graph also plots the investment in bonds when the conditional expectation of next period's marginal utility of consumption is used. This conditional expectation is calculated using the policy function for consumption and bondholdings that are implied by the 15th-order approximation. The parameter of relative risk aversion is equal to 3 and the borrowing constraint parameter is equal to 0.2. The solution using the 7th order polynomial corresponds with solution EXP4Q101 and the solution using the 15th-order polynomial corresponds with solution EXP4Q101*. Both are described in more detail in Section 3.4.]

implied by the 15th-order approximation. The graph of the conditional expectation using the consumption function that is implied by the seventh-order approximation is very similar. Notice that the conditional expectation is flat for low wealth levels. This is because, at these wealth levels, the agent is constrained and an increase in his or her wealth level does not change the amount of next period's beginning-of-period debt or the realizations of next period's consumption. As documented in the graph, the smooth polynomials have a hard time approximating the kink in the function being approximated. The effect of the inaccuracy on the policy functions is minor, however, because the function $\Psi(s, \delta^\Psi)$ is only used to determine the agent's decisions when he is not constrained. This becomes clear in Figure 3, which plots the bond function implied by the 7th- and 15th-order approximation as well as the bond function implied by the actual values of the function being approximated. As documented in the graph, the approximation errors are small, especially relative to the large changes in the level of bondholdings that occurs in this type of economy.

4.2. Accuracy of Bond Price and Transition Functions

In this section, I discuss the accuracy of the bond price function and the transition functions, taking the individual policy functions as well as the functional

form of the cross-sectional distribution as given. I focus on the baseline version of the model and analyze the accuracy of numerical solutions EXP2Q500 and EXP4Q101. Numerical solution EXP2Q500 is based on the assumption that agents' bondholdings are distributed as a second-order exponential, and numerical solution EXP4Q101 is based on the assumption that the cross-sectional distribution is a fourth-order exponential. Because the mean of the cross-sectional distribution of beginning-of-period bondholdings is equal to zero, the dimension of $\bar{s}$ is equal to 2 when numerical solution EXP2Q500 is used and is equal to 4 when EXP4Q101 is used. For this exercise, I take the functional form of the cross-sectional distribution as given. Approximations EXP2Q500 and EXP4Q101 should, therefore, be considered as solutions to different economies. In one economy, bondholdings are distributed as a second-order exponential, and in the other, bondholdings are distributed as a fourth-order exponential. In Section 4.4, I look at how accurate these two assumptions about the cross-sectional distribution are. Both numerical solutions use second-order polynomials to approximate the bond price and the transition function.

The most important determinant of the bond price in this economy is the current growth rate of the aggregate endowment. This variable, however, can take on only two values and both are elements of the grid used to solve for the approximations. In contrast, the variance of beginning-of-period bondholdings can take on a continuum of values but only five different values are included on the grid. To analyze the accuracy of the approximation, I compare the approximating bond price function $\Theta(\bar{s})$ with the equilibrium value of the bond price at a range of values for the variance. The other elements of $\bar{s}$ are kept fixed.¹⁹ Similarly, I compare the value of the approximation $\Phi(\bar{s})$ with the actual value of next period's variance for a range of values for this period's variance. If one takes the individual policy functions and the cross-sectional distribution as given, then the only approximation error made in calculating the actual values is due to errors in the numerical integration procedure. The number of quadrature nodes is chosen to be high enough to ensure that this error can be ignored.

The results for the two numerical solutions EXP2Q500 and EXP4Q101 are reported in Figures 4 and 5, respectively. Each graph plots the approximation of the bond price and the actual bond price for a certain amount of variance for the beginning-of-period cross-sectional distribution of bondholdings. It also plots the fitted and actual value for next period's variance. As documented in Figure 4, the results are excellent when the cross-sectional distribution is distributed as a second-order exponential. From Figure 5, it becomes clear that it is more difficult to approximate the bond price when the cross-sectional distribution is a fourth-order exponential. Approximation errors are still small, however. For the values used to construct Figure 5, the maximum approximation error for the bond price is 0.06%. Note that the finding that the approximation errors are smaller when the cross-sectional distribution is a second-order exponential is not very interesting if the actual distribution is not a second-order exponential. This issue is discussed in more detail in Section 4.4.

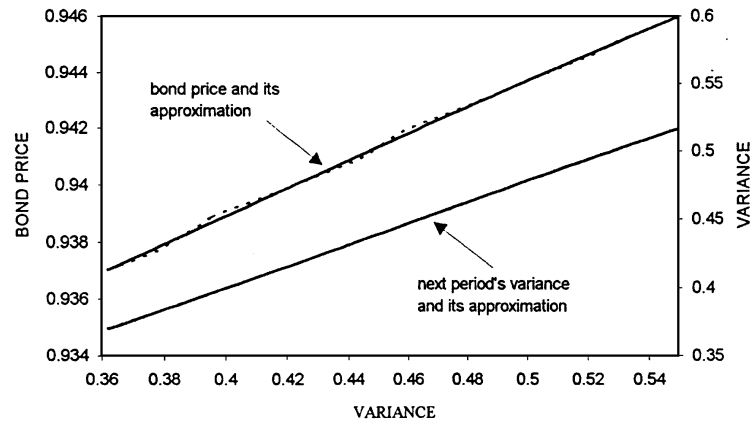


FIGURE 4. Aggregate policy functions when distribution is a second-order exponential. [This graph plots the approximation for the bond price $\Theta(\cdot)$ and the bond price at which the bond market is in equilibrium for different values of this period's variance. The graph also plots next period's variance of the cross-sectional distribution of bondholdings using both the approximation $\Phi(\cdot)$ and the actual value. The distribution of bondholdings is assumed to be a second-order exponential. See equation (8). Numerical solution EXP2Q500 is used to calculate the approximating functions. The parameters of the baseline version of the model are used. Also, the growth rate of the per-capita endowment is equal to the recession value.]

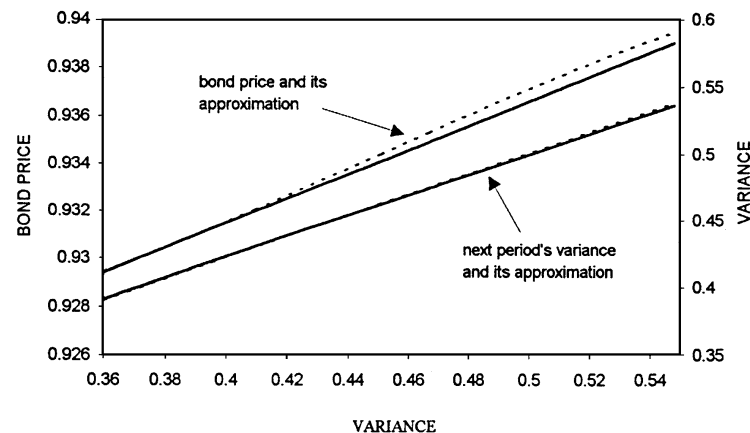


FIGURE 5. Aggregate policy functions when distribution is a fourth-order exponential. [This graph plots the approximation for the bond price $\Theta(\cdot)$ and the bond price at which the bond market is in equilibrium for different values of this period's variance. The graph also plots next period's variance of the cross-sectional distribution of bondholdings using both the approximation $\Phi(\cdot)$ and the actual value. The distribution of bondholdings is assumed to be a fourth-order exponential. See equation (8). Numerical solution EXP4Q101 is used to calculate the approximating functions. The parameters of the baseline version of the model are used. Also, the growth rate of the per-capita endowment is equal to the recession value and the values of the third and fourth moment are set equal to the value at the middle grid point.]

4.3. Accuracy of Numerical Integration Procedures

In this section, I discuss the accuracy of the numerical integration techniques used. Numerical integration techniques are used to calculate the average demand for bonds in the procedure to calculate the equilibrium bond price and to calculate the moments of next period's cross-sectional distribution of bondholdings in the procedure to calculate the transition functions. Quadrature techniques approximate $\int_a^b h(x)w(x) dx$ with a finite sum $\sum_{i=1}^n w_i h(x_i)$, where the x_i are the quadrature nodes and the w_i are the quadrature weights.²⁰ A Gaussian quadrature method is a global approach that assumes that the function $h(x)$ is a polynomial of degree $2n - 1$ over the range $[a, b]$. Simpson quadrature is local in nature. It divides the range $[a, b]$ in $(n - 1)/2$ intervals, each with three nodes, and assumes that the function is quadratic over each interval. I show in this section that, because of the kink in the demand for bonds, global quadrature methods are not appropriate and obtain an accurate approximation only with a large number of quadrature nodes.

To analyze the accuracy of the Hermite–Gaussian quadrature procedure, I repeat the exercise in the last section using 30 instead of 500 Hermite quadrature nodes to calculate a solution to the bond price function $\Theta(\bar{s})$. The results using 200 and 500 quadrature nodes are very similar. I therefore consider the results using 500 nodes as an accurate solution. The results are reported in Figure 6. The figure presents the equilibrium bond price and the second-order approximation $\Theta(\bar{s})$ for a range of values for this period's variance. Both are calculated using 30 Hermite quadrature nodes. For comparison, the graph plots the two corresponding curves from Figure 4 that are calculated using 500 nodes. As documented in the graph, there is a big difference between the bond price and its second-order approximation when only 30 Hermite quadrature nodes are used. The difference between the second-order approximation of the bond price when only 30 quadrature nodes are used and the results when 500 nodes are used, however, is a lot smaller. This suggests that taking a higher-order approximation to the bond price function would most likely give a worse approximation, i.e., a relationship that is more different than the linear relationship found with 500 quadrature points. This exercise demonstrates that, to deal with the nondifferentiability of the policy function, a large number of quadrature points is needed to obtain an accurate approximation of the bond price.

Results not reported here show that fewer nodes can be used when Simpson quadrature is used. A reduction of the number of nodes from 101 to 51, for example, has hardly any effect on the result. The explanation is relatively simple. From Figure 3, it is clear that low-order polynomials should provide a good local approximation for the agents' savings decision. Simpson quadrature, therefore, will work well with the possible exception of the area around the kink. Gaussian quadrature procedures use a global approximation. It may, therefore, provide a poor approximation of the integral even over the flat part of the function because the function can “wobble” between the grid points.

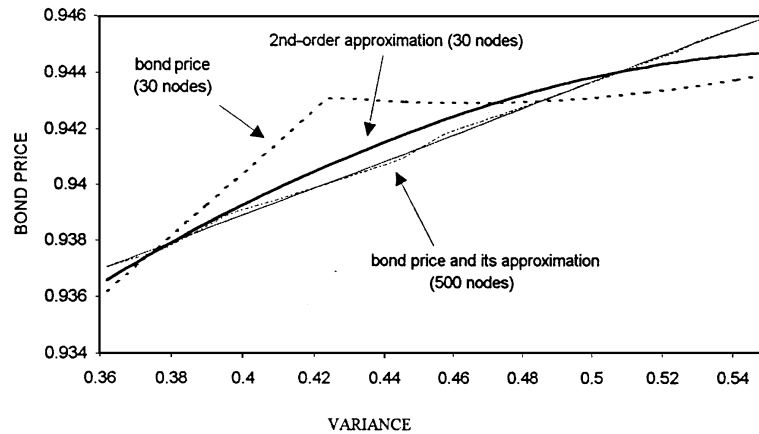


FIGURE 6. Accuracy of quadrature procedures. [This graph plots the approximation for the bond price $\Theta(\cdot)$ and the bond price for which the bond market is in equilibrium for different values of this period's variance. The solid curves are based on Hermite Gaussian quadrature with 30 quadrature nodes and the thin curves are based on Hermite Gaussian quadrature with 500 nodes. The distribution of bondholdings is assumed to be a second-order exponential. See equation (8). Numerical solutions EXP2Q30 and EXP2Q500 are used to calculate the solutions. The parameters of the baseline version of the model are used. Also, the growth rate of the per-capita endowment is equal to the recession value.]

4.4. Accuracy of Approximation of Cross-Sectional Distribution

In Section 4.2, it was shown that if the cross-sectional distribution of bondholdings is assumed to be a second-order exponential, then low-order approximations of the bond price and the transition functions are very accurate. The same result is found when the distribution is assumed to be a fourth-order exponential. The big question, though, is whether the cross-sectional distribution is indeed distributed as a low-order exponential. A researcher can, of course, start a simulation with any particular type of distribution, but it is not clear whether this distribution assumption will be satisfied in future periods. In Section 4.4.1, I describe an experiment to check the accuracy of using a finite-order exponential to approximate the cross-sectional distribution. Because this is such an important question in this paper, I consider several versions of the model. In Section 4.4.2, I report the results using the baseline version of the model in which idiosyncratic shocks are i.i.d. r.v.'s. When the idiosyncratic shocks are i.i.d., then the cross-sectional distribution of beginning-of-period bondholdings of the low-income types is equal to the distribution of the high-income types. In Section 4.4.3, I discuss several extensions of the model, including the case of serially correlated shocks.

4.4.1. Description of the experiment. To analyze the accuracy of the approximation of the cross-sectional distribution, I perform the following simulation experiment in which 100,000 agents are used to approximate a continuum of

agents.²¹ The question is whether the bond price function $\Theta(\bar{s})$ and the transition function $\Phi(\bar{s})$ correspond to the behavior of the bond price and the cross-sectional distribution in this economy. The bond price and the cross-sectional distribution are determined as follows: In period 0, the beginning-of-period bondholdings of these 100,000 agents are assumed to be distributed as a second-order exponential. A random-number generator is used to determine the growth rate of the per-capita endowment and the idiosyncratic shock for each agent. To approximate an agent's demand for bonds, I replace the conditional expectation in equation (12) with a numerical approximation $\Psi(s)$. Equations (10) through (12) then can be used to solve for the demand of bonds as a function of the state variables and the bond price. During each period, the bond price is determined as the value that sets the aggregate demand of bonds of the 100,000 members of this economy equal to zero. Using this equilibrium bond price, I then calculate for each agent the amount of bonds carried over into the next period. By repeating this experiment, one can follow the cross-sectional distribution over time and investigate whether it can be accurately approximated with a finite-order exponential. Of particular interest is the cross-sectional distribution when the economy is at its ergodic distribution. To ensure that the economy is close to its ergodic distribution, I simulate the economy for 75 periods. For the 75th period, I compare the cross-sectional distribution in the simulated economy with its approximation.²² The results below show that a fourth-order approximation provides an accurate approximation of the cross-sectional distribution, but that a second-order approximation does not.

The question arises as to how important these deviations are for the bond price and the transition of the moments of the cross-sectional distribution. The simulation experiment described above can be used to address this question. In particular, I compare the bond price and the second moment in the simulation with the ones predicted by the aggregate decision rules. For an accurate numerical solution, these two outcomes should be very similar because the sampling error is very small when 100,000 agents are used. If they are not the same, then at least one of the generated series is not accurate.

4.4.2. Baseline version of the model. In this subsection, I consider the version of the model in which the idiosyncratic endowment shocks are distributed as i.i.d. r.v.'s that can take on two values. Numerical solutions EXP2Q500 and EXP4Q101 are used. Solution EXP2Q500 assumes that the cross-sectional distribution is a second-order exponential, and solution EXP4Q101 assumes that the distribution is a fourth-order exponential. Figure 7 plots the observed cross-sectional distribution after 75 periods when numerical solution EXP2Q500 is used to determine the individuals' demand for bonds. The graph also plots the second-order approximation of this distribution.

Figure 8 is the corresponding graph when numerical solution EXP4Q101 is used to determine the demand for bonds. The 75th period turned out to be a boom. The lowest possible value of beginning-of-period bondholdings, therefore, is equal to $1/1.047 (=b/a')$. The graphs clearly show that the cross-sectional distribution

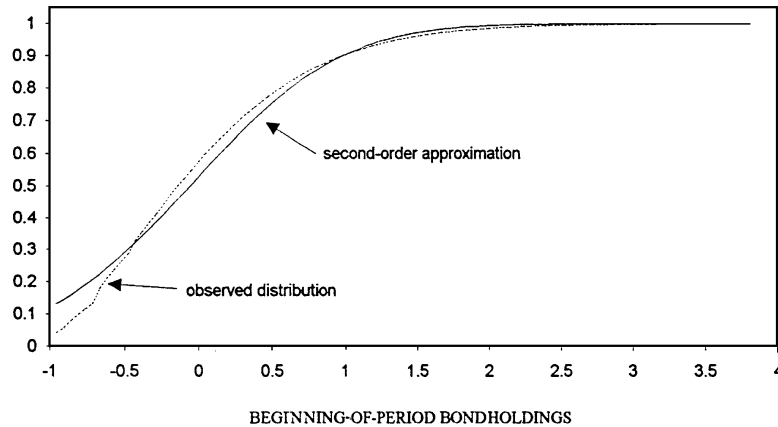


FIGURE 7. Observed cross-sectional distribution and its second-order approximation. (This graph plots the cumulative distribution function of the distribution of beginning-of-period bondholdings observed in a simulation using 100,000 agents after 75 periods. The agents' decisions are calculated using numerical solution EXP2Q500 which assumes that the cross-sectional distribution is a second-order exponential. The plotted second-order approximation has a mean and variance that are equal to the values in the simulated economy. The parameters of the baseline version of the model are used.)

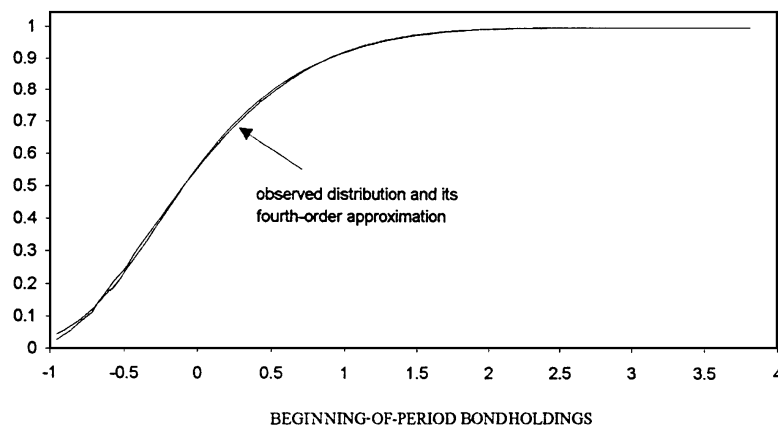


FIGURE 8. Observed cross-sectional distribution and its fourth-order approximation. (This graph plots the cumulative distribution function of the distribution of beginning-of-period bondholdings observed in a simulation using 100,000 agents after 75 periods. The agents' decisions are calculated using numerical solution EXP4Q101 which assumes that the cross-sectional distribution is a fourth-order exponential. The plotted fourth-order approximation has a mean, variance, skewness, and kurtosis that are equal to the values in the simulated economy. The parameters of the baseline version of the model are used.)

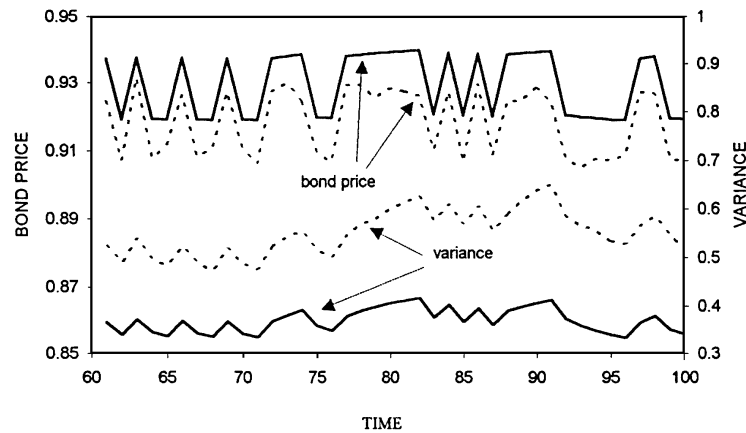


FIGURE 9. Accuracy of a second-order approximation to the cross-sectional distribution. (The dotted curves are the bond price and the variance of the bondholdings from a simulation using 100,000 agents in which the agents' decisions are calculated using numerical solution EXP2Q500. This numerical solution assumes that the cross-sectional distribution is a second-order exponential. The solid lines are the bond price and the variance calculated directly with the numerical solution EXP2Q500. The parameters of the baseline version of the model are used.)

observed in the simulated economy is not accurately described by a second-order polynomial. In particular, in the simulated sample, around 4% of the agents are at the borrowing constraint, whereas the second-order approximation would indicate that 13% are at the borrowing constraint. In contrast, the results using a fourth-order approximation are clearly more accurate.

From the results in Section 4.2, we know that the aggregate decision rules are quite accurate when the cross-sectional distribution is distributed according to a second-order exponential. The exercise above, however, showed that a second-order exponential does not provide a good approximation for the ergodic distribution of this economy. Figure 9 documents how important these deviations are for the bond price and the transition of the moments of the cross-sectional distribution. In particular, the figure compares the bond price and the second moment in the simulation with the ones predicted by the aggregate decision rules. Figure 10 reports the result when a fourth-order exponential is used. As documented in Figure 9, using the wrong assumption about the cross-sectional distribution leads to a substantial differences between the bond price predicted by the aggregate policy rule and the one predicted by the individual policy rules and the simulated distribution of bondholdings. In contrast, the numerical solution is very accurate when a fourth-order approximation is used.

4.4.3. Other parameter values. In this section, I compare the accuracy of using a finite-order exponential to approximate the cross-sectional distribution

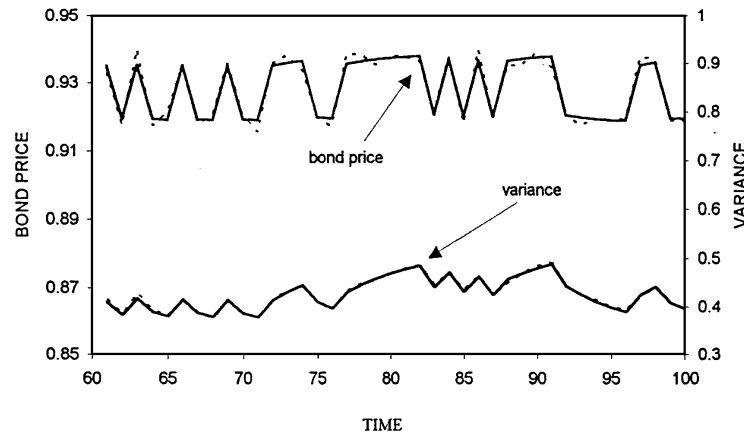


FIGURE 10. Accuracy of a fourth-order approximation to the cross-sectional distribution. (The dotted curves are the bond price and the variance of the bondholdings from a simulation using 100,000 agents in which the agents' decisions are calculated using numerical solution EXP4Q101. This numerical solution assumes that the cross-sectional distribution is a fourth-order exponential. The solid lines are the bond price and the variance calculated directly with the numerical solution EXP4Q101. The parameters of the baseline version of the model are used.)

for different versions of the model. In the first alternative version considered, the borrowing-constraint parameter is 0.2 instead of 1.0, which means that a much larger fraction of agents is at the borrowing constraint. In the second version, the idiosyncratic shock is serially correlated.

First, consider the case when the borrowing-constraint parameter is equal to 0.2. Because this value is much smaller than the value considered in Section 4.4.2, one can expect to find many more agents at the constraint. As in the preceding subsection, I compare the cross-sectional distribution observed after 75 periods with its second-order approximation. The results are reported in Figure 11. As documented in the graph, the presence of borrowing constraints implies that the cumulative distribution function is discontinuous at several values of the beginning-of-period bondholdings. These discontinuities are more pronounced for the tighter borrowing constraint. Other than missing the discontinuities, however, the approximation errors of the second-order approximation do not seem worse than the approximation errors for the case with the looser borrowing constraint. In fact, the second-order approximation now provides an accurate approximation of the fraction of agents at the constraint.²³

Finally, I consider the case in which the idiosyncratic shock is serially correlated. In this case, the distribution of beginning-of-period bondholdings across the low-income types will differ from the distribution of beginning-of-period bondholdings across the high-income types. The results for this case are reported in Figure 12. As documented in the graph, the approximation errors of the

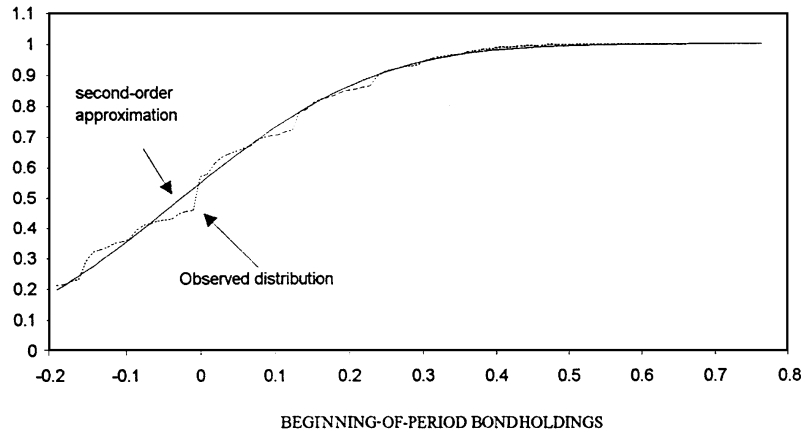


FIGURE 11. Observed cross-sectional distribution and its second-order approximation (tight borrowing constraint). (This graph plots the cumulative distribution function of the distribution of beginning-of-period bondholdings observed in a simulation using 100,000 agents after 75 periods. The agents' decisions are calculated using numerical solution EXP2Q500 which assumes that the cross-sectional distribution is a second-order exponential. The plotted second-order approximation has a mean and variance that are equal to the values in the simulated economy. The borrowing-constraint parameter is equal to 0.2. Except for this parameter, the parameters of the baseline version of the model are used.)

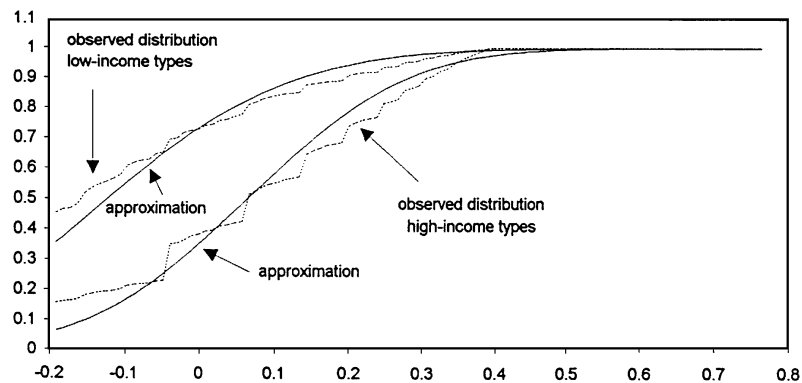


FIGURE 12. Observed cross-sectional distribution and its second-order approximation (serially correlated idiosyncratic shocks and tight borrowing constraint). (This graph plots the cumulative distribution function of the distribution of beginning-of-period bondholdings observed in a simulation using 100,000 agents after 75 periods. The agents' decisions are calculated using numerical solution EXP2Q500 which assumes that the cross-sectional distribution is a second-order exponential. The plotted second-order approximations have a mean and variance that are equal to the values in the corresponding observed distribution. The borrowing-constraint parameter is equal to 0.2 and the probability of receiving the same idiosyncratic shock is equal to 0.7412. Except for these two parameters, the parameters of the baseline version of the model are used.)

second-order approximation error are worse than in any of the other cases considered in this paper.

4.5. Sensitivity of Popular Statistics to Approximation Errors

In Section 4.4.3, it was shown that a numerical solution that used a second-order exponential to approximate the cross-sectional distribution is not capable of accurately describing the actual cross-sectional distribution. In addition, the bond price calculated by the algorithm diverged considerably from the equilibrium bond price in the simulated economy. It also was shown that a large number of quadrature nodes are necessary to obtain accurate approximation of the integrals. The question arises as to whether the properties of the model that economists are usually interested in depend very much on the particular numerical approximation used. To address this question, I compare several standard properties of the solution calculated using approximation EXP2Q30 to the corresponding properties calculated using approximation EXP4Q101. Numerical solution EXP4Q101 uses 101 quadrature nodes and a fourth-order exponential to approximate the cross-sectional distribution, and numerical solution EXP2Q30 uses 30 quadrature nodes and a second-order exponential to approximate the cross-sectional density. The properties of the model that I consider are several standard first- and second-order moments of consumption, savings, and the interest rate. The results are reported in Table 3. As documented in the table, only minor differences are observed. The intuition is that, in this model, most movements in the variables correspond to shocks in the idiosyncratic and per-capita endowment. Both numerical solutions capture these accurately. This claim is illustrated in Figure 13. Figure 13 plots an arbitrary realization for the bond price for the economy with the loose borrowing constraint using the two numerical solutions. As documented in the graph, the differences are minor relative to the movements in the bond price that are due to changes in the per-capita growth rate. The effect of these approximation errors on the consumption and savings decision depend on the particular value of the borrowing-constraint parameter used. In particular, the smaller the value of the borrowing-constraint parameter, the smaller the effect of these differences in the bond price on the consumption and savings decision of the agent. The graph also documents that some information about the cross-sectional distribution is required to accurately capture the behavior of the bond price. If the cross-sectional distribution of the bond price was irrelevant for the behavior of the bond price, then only two values for the bond price should be observed, and this is not the case.

4.6. Discussion of Nonlinearities in the Model

In this section, I summarize the role of nonlinearities in the model. First, I discuss nonlinearities in the demand for bonds as a function of the wealth level. Second, I discuss nonlinearities in the relationship between the bond price and the amount of

TABLE 3. Standard statistics calculated using an accurate and a less accurate approximation^a

	$\bar{b} = 0.2$		$\bar{b} = 1.0$	
	EXP2Q30	EXP4Q101	EXP2Q30	EXP4Q101
SD ($\Delta \ln C_t^j$)	0.112	0.112	0.0619	0.610
SD (b_t^j)	0.242	0.242	1.153	1.117
Correlation ($\Delta \ln C_t^j, \Delta \ln C_{t-1}^j$)	0.0826	0.0832	0.103	0.105
Correlation (b_t^j, b_{t-1}^j)	0.829	0.829	0.981	0.979
Correlation ($\Delta \ln C_t^j, y_t^j$)	0.629	0.630	0.533	0.542
Correlation (b_t^j, y_t^j)	0.823	0.823	0.356	0.368
Average (r_t)	-0.0423	-0.0424	0.0765	0.0777
SD (r_t)	0.0081	0.0087	0.0108	0.0095

^aThis table reports the moments calculated using numerical approximations EXP2Q30 and EXP4Q101. Numerical solution EXP2Q30 uses a second-order exponential to approximate the cross-sectional distribution and numerical solution. EXP4Q101 uses a fourth-order exponential. More information is given in Section 3.4. The moments are calculated using a simulation of 100,000 observations. The parameters of the baseline version of the model are used.

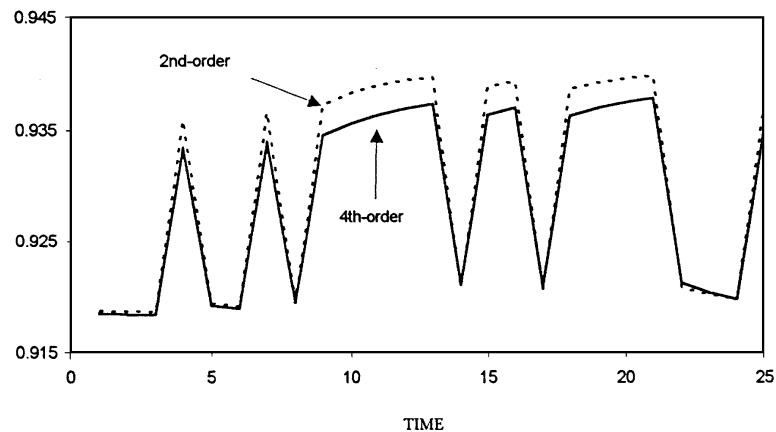


FIGURE 13. Comparing a simulation for the bond price using an accurate and a less accurate solution. (This graph plots an arbitrary simulation for the bond price using numerical approximations EXP2Q30 and EXP2Q500. The parameters of the baseline version of the model are used.)

cross-sectional wealth dispersion. Finally, I discuss nonlinearities in the response of the bond price to an aggregate shock.

In Section 4.1, it was shown that the demand for bonds displays a strong non-linearity because of the presence of borrowing constraints. As documented in Figure 3, however, for those beginning-of-period wealth levels at which agents are

not constrained, only a slightly nonlinear relationship is observed. In particular, the marginal propensity to save is slightly increasing with the wealth level.²⁴ This property of the marginal propensity to save is crucial for the relationship between the bond price and the amount of cross-sectional dispersion. Consider an economic model in which the marginal propensity to save does not depend on the wealth level. In such a model, a redistribution of wealth among the agents should have no effect on the bond price. That is, the amount of cross-sectional dispersion should have no effect on the bond price. In contrast, in the economy considered in this paper, there is a positive relationship between the bond price and the amount of cross-sectional dispersion. The results for the individual policy functions indicate that this is mainly due to the presence of borrowing constraints. Quantitatively, changes in the amount of cross-sectional dispersion are important for changes in the bond price. Suppose that in the last few periods the economy was in a recession and that the current realization of the aggregate growth rate is equal to the high value. The high growth rate immediately reduces the amount of cross-sectional dispersion in the beginning-of-period bondholdings relative to the per-capita endowment. Note that this reduction of the amount of cross-sectional dispersion is not just a rescaling of bondholdings. If each agent's endowment level increases, then the relative wealth differences between the lenders and the borrowers really decrease. Because the marginal propensity to save is increasing with the wealth level, the reduction of the cross-sectional dispersion reduces the bond price. Den Haan (1996b) calculates that the change of the interest rate in the event described is equal to 0.87 percentage points when the borrowing-constraint parameter is equal to 0.2 and the parameter of risk aversion is equal to one. If the amount of cross-sectional dispersion had stayed the same, the increase would have been only 0.52 percentage points.

Finally, I discuss how the interest rate responds to a change in the per-capita growth rate. I consider two versions of the model. In the first version, the parameters of the baseline model are used. In the second version, I consider a specification for the aggregate growth rate that implies considerably more persistence. In particular, I assume that the probability of receiving the same realization is equal to 0.7. It is useful first to consider the relationship between the variance of bondholdings and the business cycle. In Figure 14, I plot the time path of the variance when the economy goes through a recession, then a boom, and then again a recession. The initial response is the same for both specifications of the aggregate growth rate. This is because beginning-of-period bondholdings are predetermined and are expressed relative to the per-capita endowment. The response in the subsequent periods is very different. In the economy with a persistent aggregate growth rate, the variance comes back to almost the same level. In the economy with a less persistent specification, the variance continues to decline. In Figure 14, I plot the time path for the bond price. In the economy with a persistent driving process, the bond price adjusts almost instantaneously to its new level. In contrast, in the economy with a less persistent driving process, most of the adjustment takes place in the first period and it takes several periods for the bond price to stop changing. There is some intuition behind these results. Consider the experiment in which

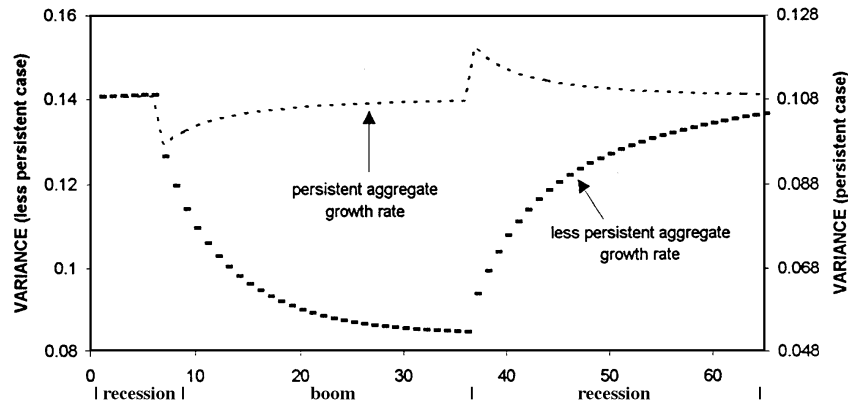


FIGURE 14. Cross-sectional variance and the business cycle. (This graph plots the time path for the cross-sectional variance of beginning-of-period bondholdings relative to the per-capita endowment when the aggregate growth rate goes through the indicated states. In the persistent specification, the probability of receiving the same realization of the aggregate shock is equal to 0.7 and in the less persistent specification it is equal to 0.5473. The other parameters are equal to the parameters of the baseline version. The model is solved using solution EXP4Q101.)

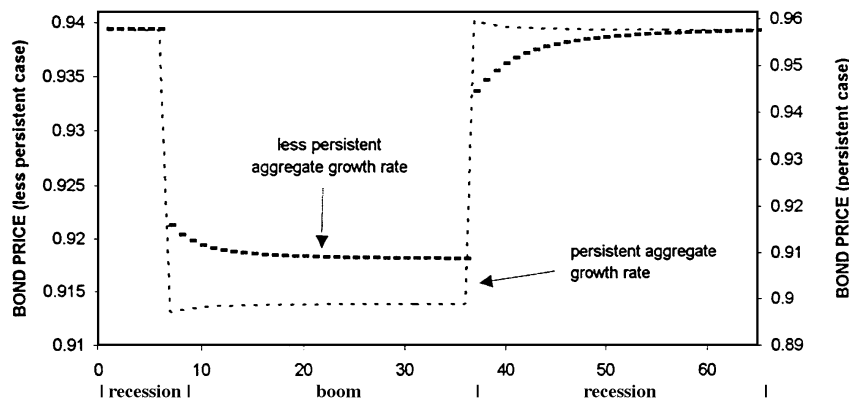


FIGURE 15. Bond price and the business cycle. (This graph plots the time path for the bond price when the aggregate growth rate goes through the indicated states. In the persistent specification, the probability of receiving the same realization of the aggregate shock is equal to 0.7, and in the less persistent specification, it is equal to 0.5473. The other parameters are equal to the parameters of the baseline version. The model is solved using solution EXP4Q101.)

the gross growth rate of the per-capita endowment is multiplied by a constant λ when the parameter of relative risk aversion is equal to 1. It is not hard to show that the variance of the beginning-of-period bondholdings relative to the per-capita endowment does not change.²⁵ The effects of this permanent change are similar to those of a persistent change in the per-capita growth rate considered in Figures 14

and 15 *when* agents perceive the change to be persistent. In the baseline version of the model, however, there is very little serial correlation in the aggregate growth rate. Agents, therefore, do not perceive a change in the aggregate growth rate to be permanent when they choose their bondholdings, and the effects on the bond price is a lot smaller. If next period's growth rate turns out to be high again, then the variance will decrease further.

5. CONCLUDING REMARKS

This paper has shown that models in which the cross-sectional distribution of agents' characteristics is an endogenous state variable can be solved accurately using commonly available computers. The model considered in this paper is relatively simple, but when faster computers become available, the algorithm easily can be extended to solve more complicated models. For example, it should be possible to incorporate a larger number of random shocks with continuous support or to allow for the investment in stocks.

There are always several choices a researcher has to make in numerically solving a model. What is best, of course, depends on the particular model being solved. In the model solved in this paper, the nondifferentiability caused by the borrowing constraint makes numerical techniques that use a sequence of local approximations more accurate than techniques using global approximations. Simpson quadrature, for example, was more accurate than Gaussian quadrature. Another important result from this paper is that some information about the cross-sectional dispersion is necessary to accurately describe the behavior of bond prices. The amount of information needed, however, depends on the purpose at hand. The paper showed that to accurately calculate standard statistics, such as the standard deviation of interest rates, little information is needed. To accurately describe the changes in the cross-sectional dispersion in response to aggregate shocks, higher-order moments are required.

NOTES

1. Examples in macroeconomics are asset pricing models with uninsurable individual risk [cf. Aiyagari and Gertler (1991), Heaton and Lucas (1992, 1996), Telmer (1993), Lucas (1994), Rios-Rull (1995)], overlapping generations models [cf. Rios-Rull (1995, 1996)], growth and investment models with different amounts of technological progress embodied across capital commodities [cf. Caballero and Hammour (1996) and Cooley et al. (1995)], entry and exit models [cf. Campbell (1995)], and Ss-type models in which there is a distribution of capital, employment, or money holdings [cf. Fisher and Hornstein (1995)].
2. See, for example, Aiyagari and Gertler (1991).
3. See, for example, Telmer (1993), Lucas (1994), and Heaton and Lucas (1996).
4. See Rios-Rull (1996).
5. Den Haan (1996b) documents the importance of this limitation by comparing a two-agent economy with the corresponding economy with a continuum of agents.
6. See, for example, Judd (1991) and Press et al. (1992).
7. Judd and Gaspar (1995) discuss techniques to deal with a large but finite dimensional state space without reducing the dimension of the state space.

8. From now on, the superscript indicating the agent is suppressed.
9. Note that one coefficient of the density is given by the restriction that the density has to integrate to one.
10. Note that the introduction of the maximum operator inside the integral relocates the mass that is below $\bar{b}/a$ at $\bar{b}/a$. The integrals are calculated numerically. For a second-order exponential, the density corresponds with that of a normal random variable, and one can apply Hermite Gaussian quadrature using a linear transformation of the variables and the appropriate adjustment of the Gauss–Hermite abscissas and weights. [Cf. Judd (1991).] In Section 4.3, I show that accurate approximations require a large number of quadrature nodes because the function inside the integral is not smooth. Using Simpson quadrature, however, accuracy is obtained with a much smaller number of nodes.
11. Or any other probability distribution function.
12. See Judd (1992) and Christiano and Fisher (1994) for the motivation of the Chebyshev approach used here.
13. In Section 4.4, I consider a version of the model in which the idiosyncratic shock has a normal distribution. In this case, Hermite Gaussian quadrature is used to numerically calculate the integral.
14. When the borrowing constraint is binding.
15. The Chebyshev polynomials also can be used when the arguments are bigger than 1 or smaller than -1 . The problem of using higher-order approximations, however, is that the behavior of the function may be highly irregular outside the range of grid points considered. In fact, the behavior can be irregular enough for the solution to be ill-defined. To avoid this problem, I impose the constraint that the agent cannot save more than the upper bound. In none of the simulated economies, each with a draw of 100,000 observations, is this constraint ever binding. Moreover, none of the results reported in this paper are sensitive to changes in this upper limit.
16. Numerical solution EXP4Q101 is used.
17. The other elements of s are kept fixed. In particular, the agent receives the low-income endowment, the aggregate growth rate is equal to the recession value, and the three moments of the cross-sectional distribution are set equal to the middle grid point.
18. Agents are allowed to borrow 20% of the per-capita endowment.
19. In particular, the economy is assumed to be in a recession. For the experiment that tests the accuracy of EXP4Q101, the third and fourth moment are set equal to the values corresponding to the middle grid point. Similar results are obtained at other values for these three variables.
20. See, for example, Judd (1991).
21. Using 300,000 agents leads to similar results.
22. Although the cross-sectional distribution changes over time in response to aggregate shocks, the results in this section are not sensitive to the particular period chosen.
23. That is, 22.5% of the agents are at the constraint in the observed sample and 19% of the agents are at the constraint according to the approximation.
24. Carroll and Kimball (1996) show that under very general conditions the marginal propensity to save should be increasing in the wealth level.
25. The new solution for b_{t-1} would be equal to λ times the old solution for b_{t-1} which means that b_{t-1}/a_t does not change.

REFERENCES

- Aiyagari, S.R. & M. Gertler (1991) Asset returns with transactions costs and uninsured individual risk. *Journal of Monetary Economics* 27, 311–332.
- Caballero, R. & M. L. Hammour (1996) On the timing and efficiency of creative destruction. *Quarterly Journal of Economics* 111, 805–852.
- Campbell, J. (1995) Entry, Exit, Technology, and Business Cycles. RCER working paper 407, University of Rochester.
- Carroll, C.D. & M.S. Kimball (1996) On the concavity of the consumption function. *Econometrica* 64, 981–992.

- Christiano, L.J. & J.D.M. Fisher (1994) Algorithms for Solving Dynamic Models with Occasionally Binding Constraints. Working paper series WP-94-6, Federal Reserve Bank of Chicago.
- Cooley, T.F., J. Greenwood & M. Yorukoglu (1995) The Replacement Problem. Manuscript, University of Rochester.
- Deaton, A. (1991) Saving and liquidity constraints. *Econometrica* 59, 1221–1248.
- den Haan, W. J. (1996a) Heterogeneity, aggregate uncertainty and the short term interest rate. *Journal of Business and Economic Statistics* 14, 399–411.
- den Haan, W.J. (1996b) Understanding Equilibrium Models with a Small and a Large Number of Agents. Working paper 5792, National Bureau of Economic Research.
- Duffie, D., J. Geanakoplos, A. Mas-Colell & A. McLennan (1994) Stationary Markov equilibria. *Econometrica* 62, 745–782.
- Fisher, J.D.M. & A. Hornstein (1995) (S, s) Inventory Policies in General Equilibrium. Institute for Empirical Macroeconomics discussion paper 104, Federal Reserve Bank of Minneapolis.
- Heaton, J. & D.J. Lucas (1992) The effects of incomplete insurance markets and trading costs in a consumption-based asset pricing model. *Journal of Economic Dynamics and Control* 6, 601–620.
- Heaton, J. & D.J. Lucas (1996) Evaluating the effects of incomplete markets on risk sharing and asset pricing. *Journal of Political Economy* 104, 433–487.
- Judd, K.L. (1991) Numerical Methods in Economics. Manuscript, Hoover Institution, Stanford University.
- Judd, K.L. (1992) Projection methods for solving aggregate growth models. *Journal of Economic Theory* 58, 410–452.
- Judd, K.L. & J. Gaspar (1995) Method for Solving Large-Scale Rational Expectations Models. Manuscript, Hoover Institution, Stanford University.
- Krusell, P. & A.A. Smith, Jr. (1994) Income and Wealth Heterogeneity in the Macroeconomy. RCER working paper 399, University of Rochester.
- Levine, D.K. (1989) Infinite horizon equilibrium with incomplete markets. *Journal of Mathematical Economics* 18, 357–376.
- Levine, D.K. & W.R. Zame (1993) Debt constraints and equilibrium in infinite horizon economies with incomplete markets. *Journal of Mathematical Economics* 26, 103–131.
- Lucas, D.J. (1994) Asset pricing with undiversifiable income risk and short sales constraints: Deepening the equity premium puzzle. *Journal of Monetary Economics* 34, 325–342.
- Magill, M. & M. Quinzii (1994) Infinite horizon incomplete markets. *Econometrica* 62, 853–880.
- Press, W.H., S.A. Teukolsky, W.T. Vetterling & B.P. Flannery (1992) *Numerical Recipes in Fortran*, Cambridge, UK: Cambridge University Press.
- Rios-Rull, J.-V. (1995) Models with heterogeneous agents. In T.F. Cooley (ed.), *Frontiers of Business Cycle Research*, 98–125. Princeton: Princeton University Press.
- Rios-Rull, J.-V. (1996) Life-cycle economies and aggregate fluctuations. *Review of Economic Studies* 63, 465–490.
- Telmer, C.I. (1993) Asset pricing puzzles with undiversifiable labor income risk. *Journal of Finance* 48, 1803–1832.